where we have slightly abused notation by allowing the symbol $\boldsymbol{R}$ to refer to both the involution matrix and the corresponding mapping $z \mapsto \boldsymbol{R}z$.

We may seek numerical methods that mimic this property, which we refer to time-reversible numerical methods. Recall from the previous chapter that a numerical method $\mathscr{G}_h$ was said to be symmetric if it satisfied

$$\mathscr{G}_{-h} = \mathscr{G}_h^{-1}.$$

It does not automatically follow that a symmetric numerical method is time-reversible, i.e. satisfies a relation inspired by (3.14), namely

$$\boldsymbol{R} \circ \mathscr{G}_h \circ \boldsymbol{R} \circ \mathscr{G}_h = \mathrm{Id}. \tag{3.15}$$

The additional condition needed to ensure (3.15) is the *affine invariance* property (or invariance with respect to the specific affine transformation $z \mapsto \boldsymbol{R}z$).

Definition 3.1 (Affine Invariance) Let

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \boldsymbol{f}(z) \tag{3.16}$$

be a given ordinary differential equation. The transformation $\tilde{z} = \boldsymbol{A}z$, where $\boldsymbol{A}$ is a non-singular square matrix, results in the modified differential equation

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \boldsymbol{A}\boldsymbol{f}(\boldsymbol{A}^{-1}\tilde{z}). \tag{3.17}$$

Let $z_0, z_1, \ldots$ be a sequence of points obtained by application of the numerical method $\mathscr{G}_h$ to (3.16). If the same method, when applied to (3.17) produces the set of points $\boldsymbol{A}z_0, \boldsymbol{A}z_1, \ldots$, then we say that the numerical method is invariant with respect to the transformation $\tilde{z} = \boldsymbol{A}z$. An affine invariant method is one which is invariant under any such transformation.

If a differential equation is time-reversible with respect to the involution $\boldsymbol{R}$, and we apply a method which is affine invariant and symmetric, then the method will be time-reversible (3.15).

All Runge-Kutta methods and Partitioned Runge-Kutta methods are affine invariant, thus, if they are also symmetric, then they preserve time-reversal symmetry.

3.6.3 Comparing Symplectic and Time-Reversible Methods

The symplectic property

$$\mathscr{F}_t'^{T} \boldsymbol{J} \mathscr{F}_t' = \boldsymbol{J}$$